

$$\max \sum_{a \in A} \sum_{c \in C} TA_{ac} * OA_{ac} + v * TA_{ac} * IA_{ac} + \sum_{b \in B} \sum_{c \in C} TB_{bc} * OB_{bc}$$

Variables:

$SA_a = 1$ if action a is selected, 0 otherwise, for $a \in A$

$TA_{ac} = 1$ if action a targets combatant c , for $a \in A$

$SB_b = 1$ if bonus action b is selected, 0 otherwise, for $b \in B$

$TB_{bc} = 1$ if action b targets combatant c , for $b \in B$

Parameters:

OA_{ac} : Threat_out if an action a targets combatant c for $a \in A$

IA_{ac} : Threat_in if an action a targets combatant c . This includes potential AoOs, maybe a fraction of the combatant's dmg threat output, for $a \in A$ where $|A| = |\text{feasibleactions}|$, $|C| = |\text{combatants}|$

OB_{bc} : Threat_out if a bonus action b targets combatant c for $b \in B$

Constraints:

$$SA_a \leq \sum_{c=1}^{|C|} TA_{ac}, \forall a \in A$$

This ensures that if an action is selected then it has at least one target. This may not be necessary since we're maxing.

Where:

$|A| = |\text{feasibleactions}|$

$|B| = |\text{feasiblebonusactions}|$

$|C| = |\text{combatants}|$